

— START SMARTER

Welcome

Welcome. You're about to learn how the most powerful tool of your lifetime actually works. Not five quick tips. Not yesterday's hype. The real machinery.

WHY WE BUILT THIS

We're Luke and Nate, and yeah, we're still in high school. This started at our kitchen table. AI is in every headline now, and it's going to be in every career, including ours. We're the ones who'll be applying for those jobs, so we'd rather understand it now than scramble to catch up later.

We went looking for something good to learn from, but most of it was shallow, focused on tools, or already out of date. So we built our own. The two of us did the building, while our dad taught us the AI itself and showed us how to actually put a course together.



Building it is what showed us the bigger point. AI is already everywhere in your life, and it'll define the career you haven't started yet. And here's the part nobody tells you: the better AI gets, the more it pays to be the person who actually understands it. The gap between people who get how it works and people who just type into the box is only going to get wider, and **which side you land on is up to you**. That edge has a name, and it's the most important AI skill there is: **Be Smarter Than the Tool**.

You'll even catch us in a few of the examples later, usually wearing the jerseys of the greatest hockey team in the world.

Here's the path: six moves, in order.

Each one builds on the last. It's the route this whole course takes.

1

Understand

How the machine works.

2

Work

Use it to get a better answer.

3

Check

Decide if what it gave you is true and useful.

4

Avoid

Spot the traps in the answers, in you, and in what you see.

5

Embrace

See what's coming, without panic or hype.

6

Build

Turn AI into your personal edge.

To start, six honest questions. Yes or no. No score. No wrong answer. You'll probably answer 'No' to at least a few. That is the point.

 TRY IT**Do you know how AI does this?**

01

Do you know why AI writes a sentence one word at a time, like a person would?

Yes

No

02

Do you know how ChatGPT, Claude, and Gemini seem to know something about everything?

Yes

No

03

Do you know why every AI tells you it makes mistakes and to double-check its answer?

Yes

No

04 Do you know which skills keep you ahead of the tool instead of behind it?

Yes

No

05 Do you know why AI gives different answers to the same question?

Yes

No

06 Do you know why even the people who built AI can't explain how it got its answer?

Yes

No

HOW LESSONS WORK

Four patterns repeat throughout the course. Spotting them tells you what each moment is asking of you. You'll need to complete the **TRY IT** and **SEE IT** activities to move to the next lesson.

TRY IT

You'll do something

Activities where you click, sort, pick, or answer. You just did one.

SEE IT

You'll watch something unfold

You'll see step-by-step how AI actually works. Less reading, more watching.

KEY INSIGHT

The principle to remember

LAB

You'll build something real

When you see this, slow down.
It's the idea the lesson is built
around.

Bigger hands-on projects with
real AI tools. You'll leave with
something you can actually use.

WHAT YOU'LL NEED

The course itself runs in your browser. Nothing to install. The labs use two outside tools, both free: **NotebookLM**, which is free with a Google account, and the free version of **ChatGPT, Claude, or Gemini**. Have those accounts ready before your first lab.

ALMOST THERE

Complete activities to continue

— START SMARTER

Why Learn AI?

Pretend you're a scribe in the year 1500. All day, you hand-copy the king's proclamations and send them out across the kingdom. Good work. Steady job. Then one morning, you hear that a guy in Mainz built a machine that prints pages a thousand times faster than you can write them. Two choices: pretend it isn't happening, or learn the new machine well enough to run it.

AI is the press. Run it, or someone else will.



IT'S ALREADY EVERYWHERE

AI isn't something you go visit. It's already in the apps on your phone, the search results you read, and the tools your first job will hand you on day one. It went from sci-fi to normal in about two years. **Most of what it does today, it will do better tomorrow.**

AI is hiding in apps you use every day

Most of the time, it's making a prediction: what you might tap, say, watch, search for, or need next.



Recommends

Predicts what you might like next.

Spotify, Netflix, TikTok



Navigation

Predicts traffic and arrival time.

Google Maps, Waze



Face recognition

Checks whether a face matches you.

Phone unlock, photo tagging



Photo search

Predicts what is inside your pictures.

Search "beach" or "dog"



Voice assistants

Turns your speech into words.

Siri, Alexa, Hey Google

AND IT'S EARLY

AI is new for everyone. This is YOUR advantage.

You're positioned to thrive in the AI future, for three reasons.

1 No one's ahead of you

The adults figuring out AI started about ten minutes before you did. Nobody has a twenty-year head start. With something this big, that almost never happens, and you're showing up right as it lands.

2 Early is a head start

The old path to real work, starting a company, shipping something people actually pay for, usually meant years of paying dues: learn the trade, climb the ladder, then maybe strike out on your own. AI collapses that runway. The leverage that used to take a decade to earn, you can reach for now.

3 Nothing to unlearn

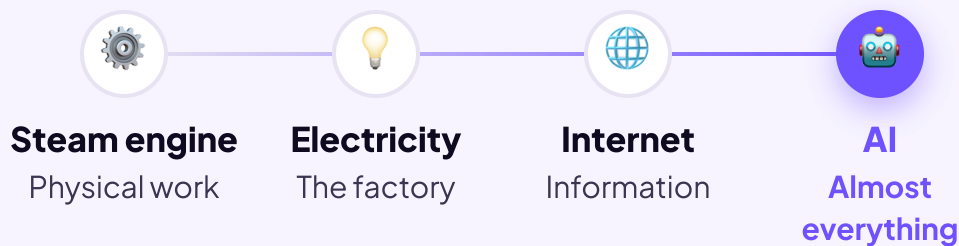
The people who got good at the old way have to undo it: break habits, give up the workflow that made them fast, and trust a tool that works nothing like what they're used to. You skip all of that. You're learning the work with AI from the start.

THIS HAS HAPPENED BEFORE

Before personal computers, becoming a designer meant years at a drafting table learning the craft by hand. Then desktop publishing showed up, and a teenager with a Mac could turn out professional work in an afternoon. The tool didn't replace skill, it shortened the distance between wanting to do the work and actually doing it. **AI is that, for almost everything.**

This is the rule, not the exception.

Each one shortened the distance between wanting to do the work and actually doing it.



We're living through the next one, and the people who learn the new tool early tend to set the terms for everyone who shows up later.

And this isn't just tech industry hype. In July 2025, the White House released an official national strategy document, *Winning the Race: America's AI Action Plan*. Here's how it describes what AI makes possible:

“

An industrial revolution, an information revolution, and a renaissance—all at once. This is the potential that AI presents.

— WINNING THE RACE · The White House, July 2025

That's not a forecast for someday. The information revolution it names, how AI moves information through your work, **is already here**.

AI won't stay in one lane. Whatever career you pick, AI will be sitting in the middle of the work. The people who learn to push it to its full potential will run circles around everyone still scratching the surface.

Remember that distance between wanting to do the work and actually doing it? AI is what shortens it now. Say you launch an online store selling hockey jerseys: where does AI close the gap for you, and where is it still on you?

🔧 TRY IT

You launch a hockey-jersey store

For each job, decide: can AI shortcut it, or is it still on you? We'll explain as you go.

01 Writing descriptions for all 40 jerseys in your catalog.

AI can shortcut it

Still on you

02 Designing a logo and a store banner before you've hired anyone.

AI can shortcut it

Still on you

03 Deciding what makes your store worth buying from instead of the ten others selling the same jerseys.

AI can shortcut it

Still on you

04 Reading three months of sales data to find which jerseys actually sell.

AI can shortcut it

Still on you

05 Replying to 50 "where's my order?" messages overnight.

AI can shortcut it

Still on you

06 Earning the trust of your first hundred customers.

AI can shortcut it

Still on you

AND IT CUTS BOTH WAYS

There's a flip side. The same tool that hands some people an edge quietly works against anyone who doesn't understand it. AI sounds sure when it's wrong, slips in answers you never asked for, and shapes what you read without ever saying so. If you can't tell when it's off, the tool is using you. Learning how it works is how you stay in charge.

 **Offense and defense, one skill.** Understanding how AI works is what turns it into your edge, and what keeps it from quietly using you. That's why you learn it now, not later.



PREVIOUS LESSON
Welcome

ALMOST THERE

**Complete activities to
continue**

— START SMARTER

What Is AI?

Artificial intelligence (AI) is software built to do things that used to need a human brain: learning from examples, spotting patterns, seeing, hearing, and using language.

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.

LEARN ONCE

 **Training**



 **Patterns**



patterns power
every answer

ANSWER · EVERY WORD

 **Probability**



 **Prediction**

Four ideas carry it, and we'll follow one example, peanut butter, through all of them.

01/4

LEARN ONCE

● Training

○ Patterns

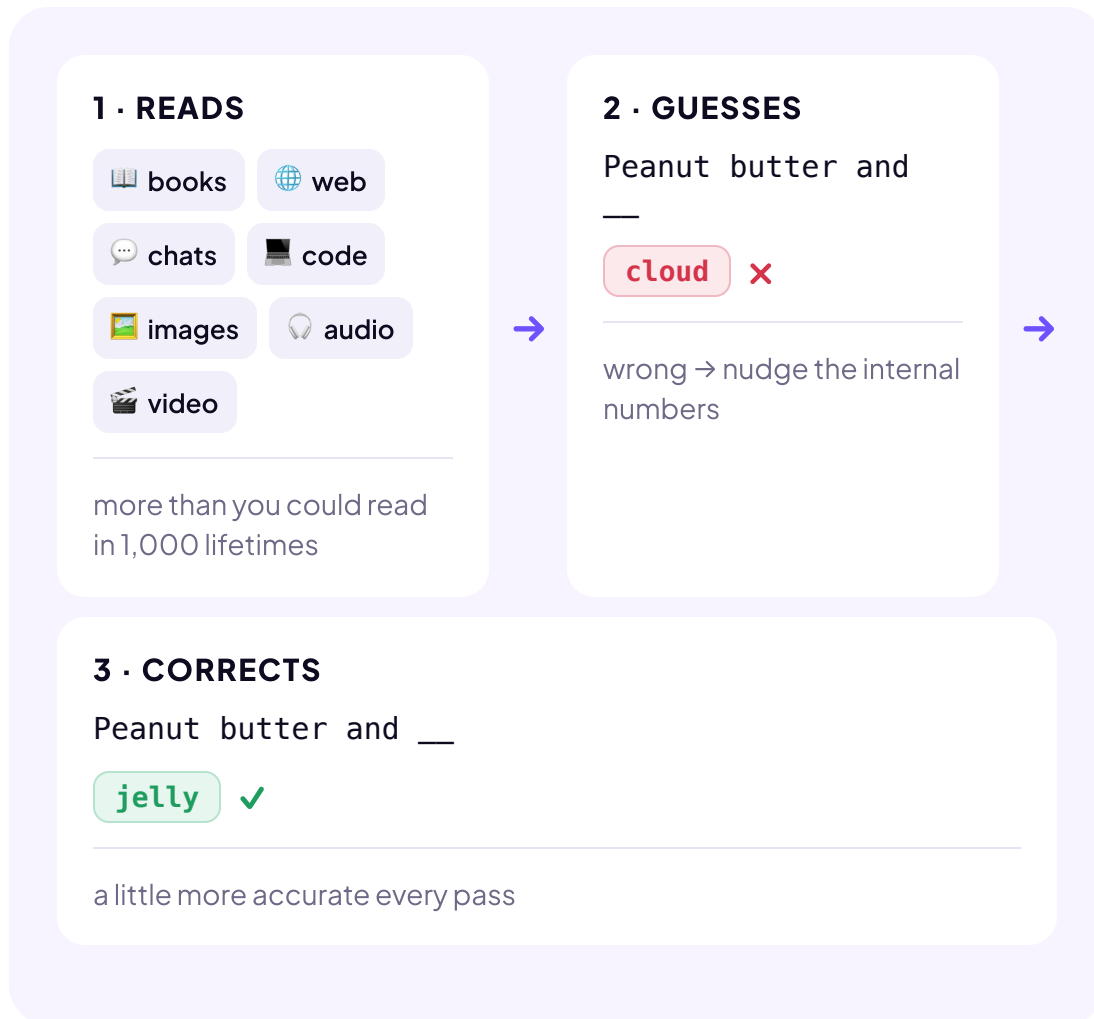


EVERY WORD

○ Probability

○ Prediction

The model **teaches itself**: guess the next word, check, and nudge its numbers when it's wrong.



Repeat that loop billions of times, and that's training.

02/4

LEARN ONCE

○ Training

● Patterns



EVERY WORD

○ Probability

○ Prediction

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

Peanut butter and _____ → **jelly**

you knew it, so does AI

PATTERNS ARE EVERYWHERE

Twinkle, twinkle, little ____ → **star**

Once upon a ____ → **time**

Better late than ____ → **never**

AI didn't memorize these. It absorbed the pattern by running through billions of examples.

03/4

LEARN ONCE ○ Training ○ Patterns



EVERY WORD ● **Probability** ○ Prediction

AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.

SAME WORD, DIFFERENT ODDS

I'd like to buy peanut butter and _____.

jelly		41%
bread		27%
bananas		16%
honey		5%

I'd like to buy a peanut butter and banana _____.

sandwich		54%
smoothie		16%
toast		9%

jelly

2%

Add the word banana, and jelly's probability drops from 41% to 2%.
The surrounding text changed everything.

04/4

LEARN ONCE

☐ Training

☐ Patterns



EVERY WORD

☐ Probability

☒ Prediction

Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next. AI works the same way, with no fixed plan for where the sentence will end up. One word, look again, the next.

1 I want to buy peanut butter and

+

jelly

2 I want to buy peanut butter and

jelly

+

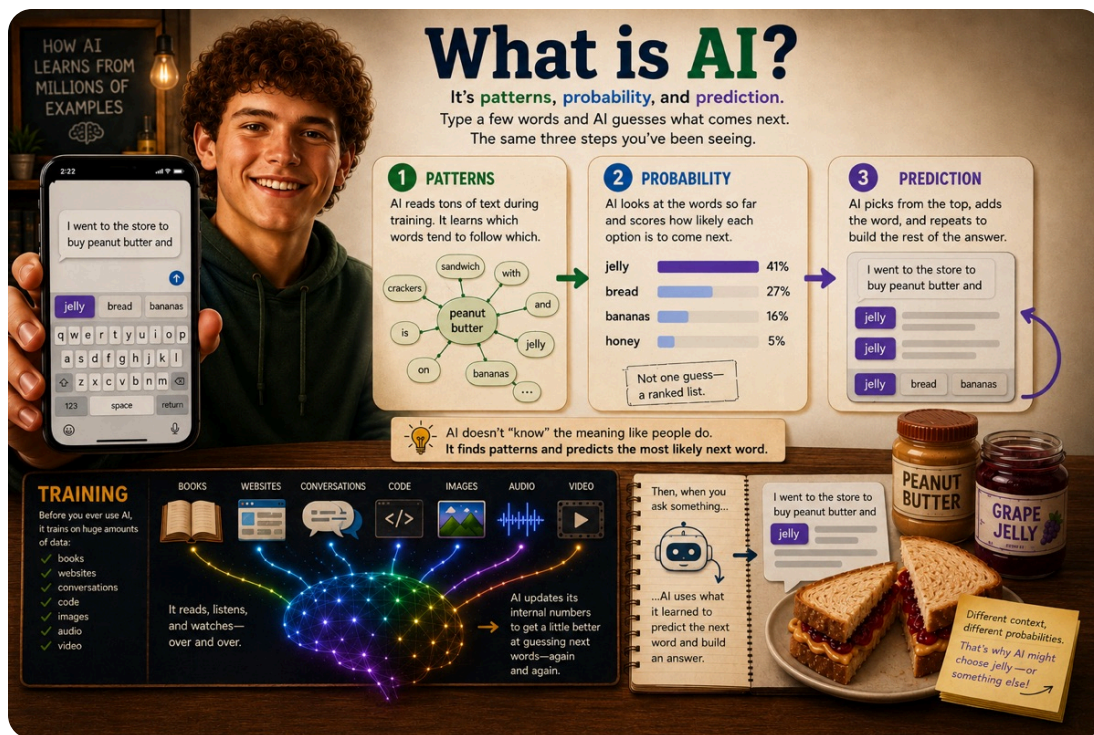
for

3 I want to buy peanut butter and jelly
for

+

lunch

A full paragraph runs this loop many times, fast enough to look like thought.



That clears up three big myths:

- AI isn't **magic** — it's math working out probabilities.
- It isn't **a person** — no thoughts, no understanding, even when it sounds like it has both.
- It isn't **a search engine** — it doesn't pull up a stored answer, it builds a new one each time.

Keep those three straight and much of the confusion falls away.

TRY IT

True or false?

One statement for each idea you just met. True or false?

01

AI learns by reading mountains of text and nudging itself every time it guesses the next word wrong.

True

False

02 There are people in a dark data center typing out the answers.

True

False

03 AI memorizes the exact answer to every question and looks it up later.

True

False

04 AI has read more text than any person could in a thousand lifetimes.

True

False

05 When AI picks the next word, it scores every option, and the surrounding text changes those scores.

True

False

06 AI knows hockey rules because it grew up in Canada.

True

False

07 AI plans out the whole sentence before it writes the first word.

True

False



PREVIOUS LESSON

Why Learn AI?

ALMOST THERE

Complete activities to
continue

— START SMARTER

Meet the Tool

You've seen how AI works: it learns patterns, then predicts an answer one word at a time. Now let's name the thing you're actually using when you open ChatGPT, Claude, or Gemini.

Two kinds of AI (and you already use both)

 SORTS & RECOMMENDS

Recommendation AI

Chooses from what already exists



Ranks the options and picks the top one

WHERE YOU MEET IT

Next show on Netflix

Next song on Spotify

Fastest route on Maps

Never makes anything new.

 CREATES NEW CONTENT

Generative AI

Makes something that didn't exist

Twin boys in highschool, who love the Dallas Stars, are building an AI course at a warm |

kitchen table.

This prompt made the picture on the welcome page.

WHAT IT CAN DO

Writes the email

Drafts the essay

Makes the image

Codes the website

When people picture AI, they mean this, and it's what this course is about.

Here's what that difference looks like on one task.

● SEE IT

Two kinds of AI, one task

THE TASK

"What movie should I watch tonight?"

See what Netflix recommends →

THE NAME GAME

You'll hear a pile of names: AI, generative AI, ChatGPT, LLM. Don't let the terminology trip you up. For the rest of this course, we'll usually say AI, and we mean the generative kind that you experience with ChatGPT, Claude, and Gemini.

There's a name for these generative AI tools: LLM, short for **Large Language Model**. Think of it like this. ChatGPT is the app you use, and the LLM is the engine under the hood doing the work.

● SEE IT

What's an LLM? Click each letter below.

LLM

🔑 **The tool, named:** The AI you'll use all course is generative AI, powered by a Large Language Model. Everything ahead is about getting more out of it.



PREVIOUS LESSON

What Is AI?

ALMOST THERE

Complete activities to
continue

— START SMARTER

Does AI Think?

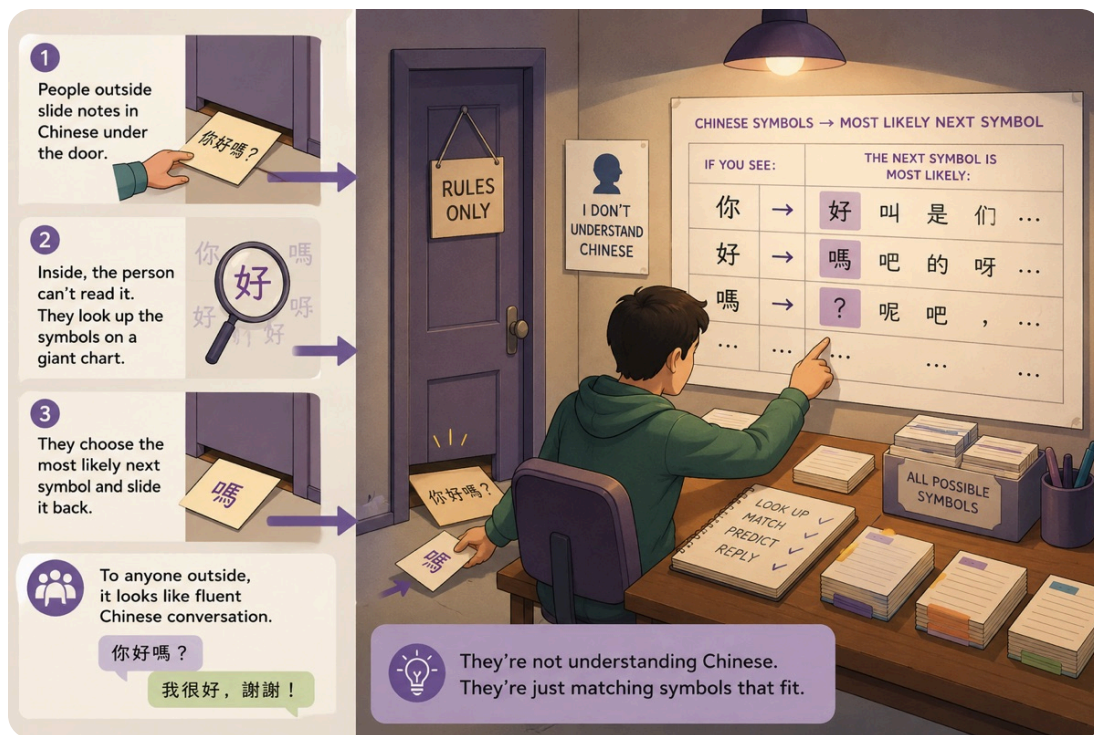
You type a question and AI types back like a person. It explains, it jokes, it says sorry when it slips up. After a while it's hard not to feel like someone is in there. So here's the big question: **Does AI actually think?** Is it alive? Is it smart? Does it feel anything?

Short answer: no, not the way you do.

IT'S IN THE NAME

We call it Artificial Intelligence, not Artificial Thinking. That was picked on purpose. Something can act intelligent, finish your sentence, explain a poem, without understanding a single thing it just said.

There's a famous way to picture this, sometimes called the Chinese Room. Imagine someone locked in a room who doesn't know a word of Chinese. Notes in Chinese come in under the door. The person can't read any of it, but they have a giant chart of which symbol tends to follow which. So they match symbols and slide back an answer that looks perfect to the people outside.



That's AI. It arranges symbols brilliantly and understands none of them. You get a perfect-looking answer with no real understanding behind it.



▶ **LISTEN** 1 min

Prefer to listen? The Chinese Room story



When you think vs. what AI does



A HUMAN MIND

When you think



A PATTERN MACHINE

What AI does instead

Understand what things actually mean



Matches patterns in mountains of text

**Draw on real
experience and
senses**



**Only ever read
about the world**

**Choose words to
make a point**



**Predicts the next
likely word**

**Actually care
whether you're
right**



**Has no stake in
being right or
wrong**

**Feel when
something's
beautiful**



**Echoes what
people call
beautiful**

**Know when
you're unsure**



**Can't tell when
it's making
things up**

None of this means AI is dumb or useless. A pattern machine this good is genuinely powerful. It just means it's not thinking. Once you see that, its weird moments start to make sense.

TRY IT

True, or Just Sounds True?

For each one, decide whether it's what AI really does or whether it just sounds that way. We'll explain as you go.

01 AI understands what your words actually mean.

True

Just sounds true

02 AI is predicting which word should come next, one at a time.

True

Just sounds true

03 AI feels bad when it gives you a wrong answer.

True

Just sounds true

04 AI builds its answers by matching patterns from a huge pile of text.

True

Just sounds true

05 AI knows when it doesn't know something.

True

Just sounds true

06 AI can sound completely sure and be completely wrong.

True

Just sounds true



PREVIOUS LESSON
Meet the Tool

ALMOST THERE

Complete activities to
continue

Does School Matter?

A lot of people your age are quietly asking the same thing. If AI can answer anything, do you even need school anymore? Why sit through years of classes when the tool spits out the answer in seconds?

Think about it this way.

Fast forward a few years. You've landed your dream job at Google, and AI sits right next to you all day, cranking out drafts, code, and plans. Your coworker at the next desk is doing the same. You each have the same job, using the same AI.

Here's the catch: your coworker, and everyone else using the same AI, gets similar answers. What makes you more valuable than your coworker? It's not what AI tells you. **It's what you already know.**

WHAT MOST PEOPLE DO NOT UNDERSTAND

In the AI era, learning is more important than it's ever been. This seems counterintuitive, but it's true. This inspired the name of this course, **Be Smarter Than the Tool**. Whatever college and career you choose, the more you learn, the more successful you'll be.

Here's what your future will look like. In almost any job, you'll have AI as your partner. So, for your dream job at Google, here's what it looks like.

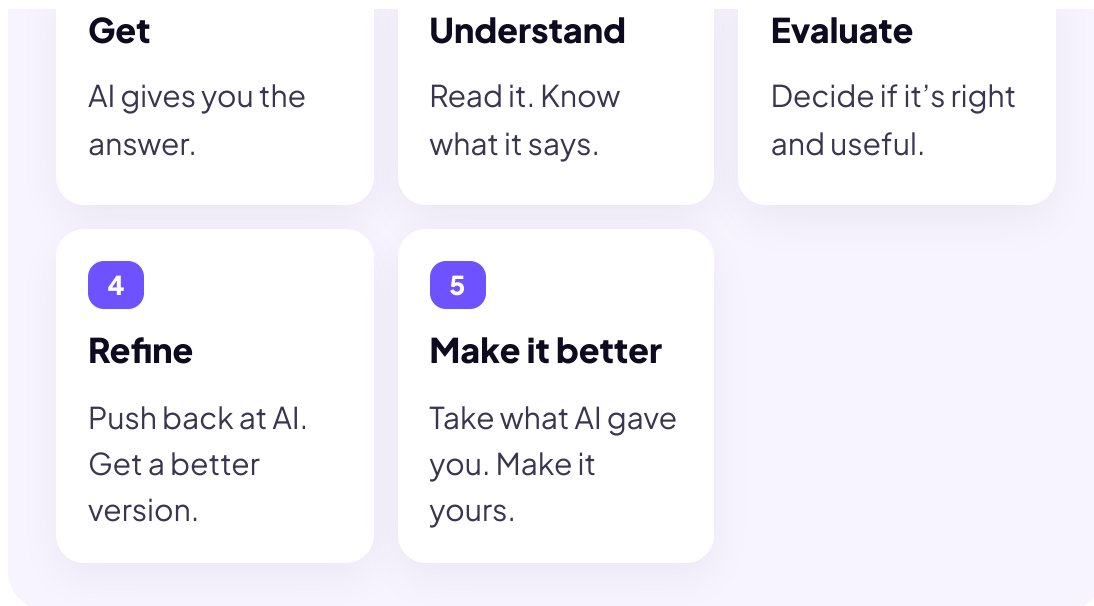
AI does step 1. You do the rest.

Steps 2 through 5 are skills. And every one of them runs on what you already know.

1

2

3



Here's the most important thing you'll learn in this course. AI will always handle step 1. It's the ultimate answer machine. Ask anything, and you'll get an answer.

But, it's not as good at steps 2 to 5. It doesn't understand what it gives you the way you do. So, how can it evaluate, refine, and make it better, when it doesn't even understand what it says?

Those steps are your domain. This is where you need to live. And it starts today, not when you work at Google. Double-down on building your knowledge and skills. And, one day you'll be the CEO at Google.



ONE TRUTH ABOUT AI

AI is programmed to give you answers. And, as you gain AI experience, you'll see that sometimes it's amazing. But other times, it's not.

For example, we used AI to help develop this class. It coded every page and every piece of interactivity you'll see. For coding, it deserves an **A+++**. It also wrote the first draft for much of the text. It was average, at best. It didn't always make points in the right order, explain things in the way we wanted, or give us the learning flow we wanted. For writing, we give it a **C-**. Much of our time was spent in the last 3 steps: Evaluate, Refine, and Make it better.

It's not that AI is broken. It's that we humans are much better at understanding, evaluating, refining, and making things better.

 TRY IT

More Knowledge. Better Results.

You and your coworker have the same job, the same AI, and the same answer back from it. Each time, one choice trusts AI's output and the other applies what you know. Pick the move that makes you more valuable than your coworker.

01

Same job as your coworker, same AI. A long-time customer is upset and threatening to leave. You both ask AI what to say, and it writes a polished, professional reply. But you know this customer. They have been with you five years, and right now they just want to feel heard, not handled.

Send AI's polished reply

Answer like you know this customer

02

You both ask AI to build a new feature, and it writes code that passes its first test and looks finished. But you remember what you learned: passing one test is not the same as holding up when a million people hit it at once.

Ship it, the test passed

Load-test it first, the way you learned

03

The hardest one. You both ask AI to write the launch email for that feature. It comes back clean, with no errors, and it reads fine. Your coworker sends theirs as is. But you know your customers. They do not care about a feature list. They care about the one problem it solves for them.

Send it, it's clean and correct

Rewrite it around what your customers actually want

 **The difference is you.** AI handles Get for anyone who types. Understand, Evaluate, Refine, and Make it better only happen if you know enough to do them. That's why you learn.



PREVIOUS LESSON

Does AI Think?

ALMOST THERE

Complete activities to continue

— START SMARTER

What You Can Control

You're hearing it constantly: AI is taking jobs. That headline used to be a prediction. Now it's news.

And jobs are just the loudest headline. AI is also reshaping who holds power, what it costs the planet to run, and what you can trust online. It's all real, it's all moving fast, and almost none of it is in your hands.

WHAT IS GOING TO HAPPEN?

Will it erase millions of jobs or create new ones? Will it wreak havoc on the environment? **Nobody knows yet, including the people making the predictions.** That uncertainty feels heavy, and part of it is fair.

Reading big AI predictions is its own skill, and you'll build it later in The Future lesson. Right now, the headline misses the only question worth asking: what's in your hands, and what isn't?

OUT OF YOUR HANDS

You don't control these

**Which jobs
change first**



IN YOUR HANDS

You do control these

**What you
actually learn**

**Headlines and
hype**



**How much the
hype gets to you**

**School and
workplace
policies**



**How hard you
work in school**

**How fast AI
improves**



**How well you use
AI**

**New laws and
regulations**



**Whether you
keep thinking for
yourself**

Most of the noise around AI is the left column. Most of the leverage in your life is the right column.

 TRY IT

In Your Hands or Not?

For each one, decide whether it's in your hands or not. We'll explain why as you go.

01 Whether the company you want to work for uses AI

Can't control

Can control

02 How worked up you get by every AI doom prediction

Can't control

Can control

03 Whether AI replaces entry-level work in your field

Can't control

Can control

04 Whether you skip the thinking and let AI do it for you

Can't control

Can control

05 Whether you build skills no one can hand you

Can't control

Can control



PREVIOUS LESSON
Does School Matter?

ALMOST THERE

Complete activities to
continue

— START SMARTER

Study with AI

Here's the good news: when it comes to studying, AI might be the most powerful tool you'll ever get your hands on. Used the right way, it doesn't just help you finish the work, it makes you genuinely sharper at the subject, and that shows up in your grades.

It's the most patient tutor you'll ever meet. It'll quiz you at 1am, never sighs when you ask again, and meets you exactly where you're stuck. Point it at your work the right way and the grades follow. The first move is choosing which kind of AI to aim at the job.

One question picks the tool: do you want to learn from materials you already have, or learn something new?

Which study tool for the job?

✨ CHATGPT · CLAUDE · GEMINI

General Tutor

Makes things from scratch.

Trained on massive amounts of data, it can explain or quiz you on almost anything.

🎯 BEST USE

You don't have the materials, or you

📖 NOTEBOOKLM

Source-grounded Tutor

Learns from your materials.

It only knows what you give it, which makes it the ultimate exam-prep machine.

🎯 BEST USE

You have the materials the test

want a concept explained a new way or extra practice your notes don't cover. Think a paper on a new topic, or something not

⚠ THE CATCH

It can occasionally hallucinate (make things up), or explain a concept differently than the way your teacher taught it and expects on the exam.

covers: class notes, screenshots, study guides, even a webpage or YouTube video. Got them? Start here.

⚠ THE CATCH

It doesn't know what you didn't give it. If your notes are missing a key concept, it can't fill in the blank for you, because it isn't allowed to look outside its binder.

START HERE

Use NotebookLM as your main study tool. The clue is in the name: it was built for learning.

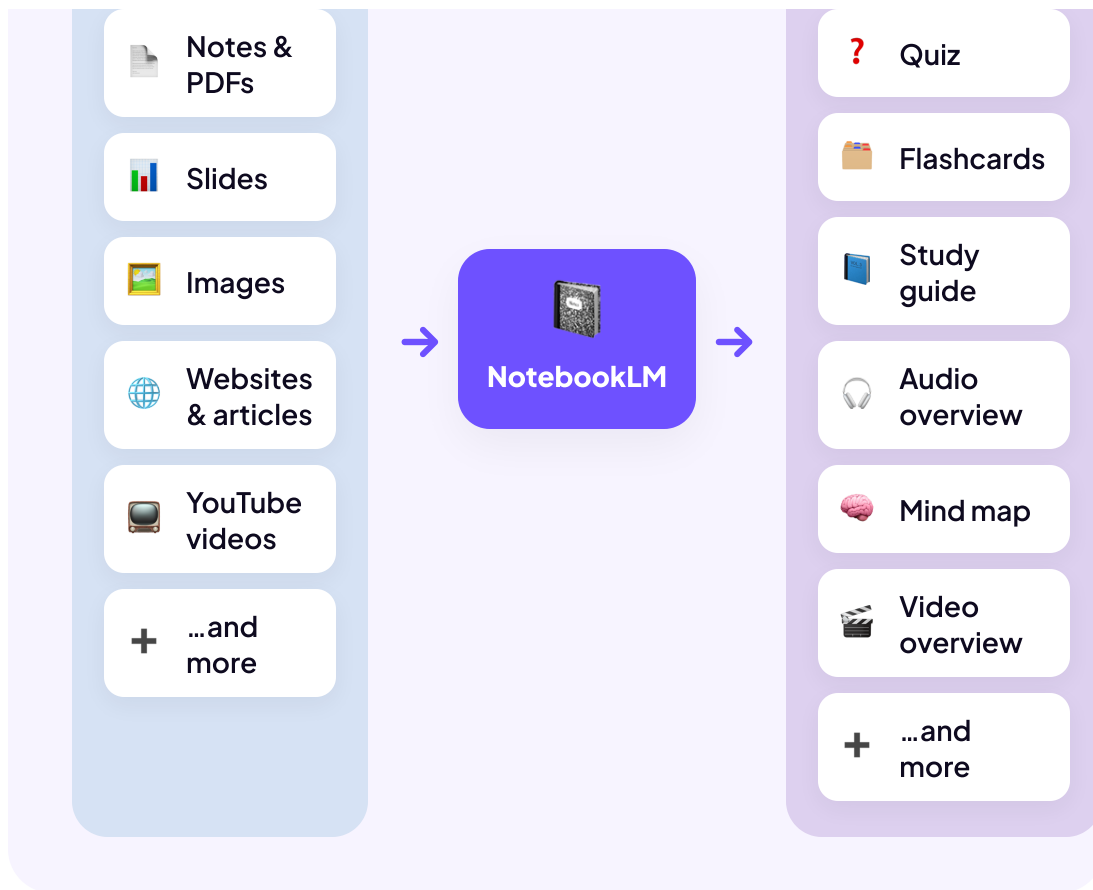
An AI app like ChatGPT or Claude can do most of the same things, but it's more difficult because they weren't designed with the goal of learning.

HOW NOTEBOOKLM WORKS

Feed it your own material and it turns that into study tools, built only from what you gave it.

WHAT YOU FEED
IN

WHAT YOU GET
OUT



BEST PRACTICES

Here's how to use NotebookLM to its full potential.

1 📁 **One notebook per subject**

Keep AP History and Chemistry in separate notebooks. Mixing sources confuses the model and gives you muddled study guides. Name each one clearly and treat it like its own binder.

2 📖 **Feed it a few angles, not one**

For a big test, give it everything your teacher handed you: your class notes (so it keeps your teacher's wording), the slides, and any video or link the teacher shared. NotebookLM reads a video's transcript and blends it with your notes, so everything it gives back stays true to your class.

3 ? Quiz yourself blind

Re-reading notes feels productive, but that's recognition, not recall. Generate a quiz, close everything, and take it cold.

"Generate a 10-question quiz from these sources only. Don't show the answers yet. Wait for mine, then grade me."

4 🗣️ Teach it back out loud

Have it build a slide deck or audio overview, then explain each part to an empty room, or pause the podcast and say it back in your own words. Wherever you stumble is exactly what you don't know yet.

5 🔍 Spot-check the citations

NotebookLM stays grounded in your sources, so it's usually right, but it can still misread a sentence. You don't need to check every line. When a fact surprises you, click its numbered bubble to jump to the exact spot in your notes.

🔑 **Pick the tool, then make it test you.** Point NotebookLM at your own notes and slides, or a general AI at a topic you don't have materials for. Either way, the win comes from producing the answers yourself, not watching it summarize. That's the difference between feeling ready and being ready.

✂️ LAB 01

Build Your Course Notebook

About 25 minutes. You'll need a laptop and a Google account.

☐ **1. Download your packet**

[Download the Start Smarter packet \(PDF\)](#) and save it to your computer. That one file is the whole first part of the course.

☐ **2. Create the course notebook**

At notebooklm.google.com, make a new notebook and name it after the course: **Be Smarter Than the Tool**. This notebook grows with you. Every part of the course will land in it.

☐ **3. Upload the packet**

Click **+ Add sources** and upload the Start Smarter PDF you just downloaded.

☐ **4. Generate a quiz to check your knowledge**

Use the prompt from this lesson: “Generate a 10–question quiz from this source only. Don’t show the answers yet. Wait for mine, then grade me.”

☐ **5. Take the quiz**

Answer all ten, then let it grade you.

☐ **6. Click one citation**

Pick one graded answer and click its numbered citation. It lands on the exact passage of the course that taught it. Make that check a habit.

☐ **7. Transfer it**

This week, make a second notebook for whichever class has your next test, and feed it your real materials: notes, slides, anything your teacher shared. The course showed you the machine; your grades are where it pays.

THE PAYOFF

**This is how
you study
every class
now.**

You just ran the whole loop on this course. Do the same for any class with a test: feed it your notes and slides, let NotebookLM quiz you, and close the gaps before exam day. That's studying that sticks, and it shows up in your grades.



PREVIOUS LESSON

What You Can Control

NEXT LESSON

Foundations

